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Rotatable camera mounting device with a rotatable disc device

Abstract

The present disclosure provides a rotatable camera mounting device with a rotatable disc device, wherein comprising: a driving device comprising a motor, a rotatable disc device connected to the motor, and a rotating device connected to the rotatable disc device, wherein the rotating device is configured to rotate circumferentially with the motor; and a rotatable disc fixing member used to fix the rotatable disc device on the rotatable camera mounting device.

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Background/Summary

FIELD

(1) The present disclosure relates to the field of photography, and more specifically, to a rotatable camera mounting device with a rotatable disc device.

BACKGROUND

(2) With the development of society, people have increased their demand for the stability, intelligence, usability, and human-computer interaction of the camera mounting device. Traditional rotatable camera mounting devices are generally used to take photos for the objects or cannot carry heavy objects, and have a single or simple function and a low level of integration. There is an urgent need for a rotatable camera mounting device suitable for human and objects. The rotatable camera mounting device can be equipped with a 3D holographic device, which can display contents such as time, pictures, animations, etc., further enhancing the entertainment; a stable skeletal structure, including multiple reinforcing beams, greatly enhancing the strength of the upper base; LED ambient light, which enhances the fun and playability of the device; an AI voice control unit, greatly improving the efficiency and scene application of the device by adjusting the rotation of the device, switching or changing lights, and turning on and off the device by voice, enhancing the interaction between people and the device; the main electrical components such as the control device and motor are all located on the lower base, which can be installed and removed without disassembling the upper base, making it easy to install and maintain the entire machine.

SUMMARY

(3) The technical problem solved by the present disclosure is the insufficient stability, intelligence, usability, and human-computer interaction of the existing rotatable camera mounting device.

(4) In view of the above and other ideas, the present disclosure is proposed.

(5) According to one aspect of the present disclosure, there is provided a rotatable camera mounting device with a rotatable disc device, wherein comprising: a driving device comprising a motor, a rotatable disc device connected to the motor, and a rotating device connected to the rotatable disc device, wherein the rotating device is configured to rotate circumferentially with the motor; and a rotatable disc fixing member used to fix the rotatable disc device on the rotatable camera mounting device.

(6) In one embodiment, wherein the rotatable disc device comprises a bearing inner ring fixed to the rotatable camera mounting device by the rotatable disc fixing member, and a bearing outer ring that rotates relative to the bearing inner ring.
(7) In one embodiment, wherein the driving device further comprises a driving wheel connected to the motor, and the bearing outer ring rotates with the driving wheel.
(8) In one embodiment, wherein the rotatable camera mounting device comprises a lower base, which is provided with a lower base frame and a third reinforcing skeleton, and the rotatable disc fixing member is fixedly installed on the third reinforcing skeleton.
(9) In one embodiment, wherein the rotatable camera mounting device further comprises: an upper base arranged opposite to the lower base, wherein, the upper base comprises a loading surface and an upper base body, the upper base body comprises an upper base frame and a first connecting member, the first connecting member is provided with a first connecting surface at one end away from the loading surface.
(10) In one embodiment, wherein the rotatable camera mounting device further comprises a second connecting member connected to the first connecting surface and connected to the bearing inner ring.
(11) In one embodiment, wherein the second connecting member is provided with a second connecting surface near the first connecting surface, and the first connecting surface is connected to the second connecting surface.
(12) In one embodiment, wherein the second connecting member is provided with a second connecting end away from the first connecting surface, and the second connecting end is connected to an upper end surface of the bearing inner ring.
(13) In one embodiment, wherein the rotatable disc fixing member is provided with a third connecting surface near one end of the bearing inner ring, and the third connecting surface is connected to a lower end surface of the bearing inner ring.
(14) In one embodiment, wherein the third reinforcing skeleton intersects with each other to form a fourth connecting end for connecting the rotatable disc fixing member.
(15) In one embodiment, wherein the rotating device comprises a first rotating unit connected to the bearing outer ring, a second rotating unit connected to the first rotating unit, and a third rotating unit rotatably connected to the second rotating unit.
(16) In one embodiment, wherein the second rotating unit is nested inside the first rotating unit.
(17) In one embodiment, wherein fasteners are provided on the first rotating unit to adjust length of the second rotating unit nested inside the first rotating unit and to fix the second rotating unit.
(18) In one embodiment, wherein one end of the first rotating unit connected to the bearing outer ring is provided with multiple connecting wings, which extend outward to form a clearance space for avoiding the second connecting end.
(19) In one embodiment, wherein the first rotating unit comprises multiple first cantilevers, the second rotating unit comprises second cantilevers corresponding to the first cantilevers, and each of the second cantilevers is nested inside the first cantilever.
(20) In one embodiment, wherein the upper base body further comprises a first reinforcing skeleton connected to the upper base frame, the first reinforcing skeleton comprising two or more first reinforcing beams, and the two ends of each of the first reinforcing beams are connected to the upper base frame.
(21) In one embodiment, wherein the upper base body further comprises a second reinforcing skeleton, wherein the second reinforcing skeleton comprises two or more second reinforcing beams, and the two ends of each of the second reinforcing beam are respectively connected to the upper base frame and the first reinforcing beam.
(22) In one embodiment, wherein the first connecting member comprises grooves that match with the first reinforcing skeleton.
(23) In one embodiment, wherein the first connecting member is provided with a first limiting

surface opposite to the first connecting surface.
(24) In one embodiment, wherein the loading surface is provided with a first opening, the first connecting member is provided with a third opening, the rotatable camera mounting device further comprises a 3D holographic device, the 3D holographic device comprising a 3D holographic device body extending into an accommodating cavity through the third opening and a 3D holographic device fixing surface, and the accommodating cavity is enclosed by the first connecting member and the first reinforcing skeleton.
(25) In one embodiment, the rotatable camera mounting device further comprises an AI voice device and a main control device, wherein the main control device is configured to create instructions and control the rotatable camera mounting device to perform actions after obtaining user voice collected by the AI voice device.
(26) In one embodiment, the AI voice device comprises a microphone device configured to capture the user's voice when the user speaks.
(27) In one embodiment, the AI voice device further comprises a speaker device configured to provide voice feedback under the control of the main control device when the microphone device acquires user voice.
(28) In one embodiment, the action includes controlling activation, deactivation, lighting mode switching, or brightness adjustment of the ambient light.
(29) In one embodiment, the loading surface is detachably connected to the upper base frame and/or the loading surface is integrally formed with the upper base frame.
(30) According to another aspect of the present disclosure, there is provided a rotatable camera mounting device, wherein comprising: an upper base, wherein, the upper base comprises a loading surface and an upper base body connected to the loading surface, the upper base body comprises an upper base frame, a first connecting member, and a first reinforcing skeleton connected to the upper base frame, the first connecting member is provided with an accommodating cavity, and one end of the first connecting member away from the loading surface is provided with a first connecting surface.
(31) In one embodiment, the loading surface is provided with a first opening.
(32) In one embodiment, the first connecting member is provided with a second opening opposite to the first opening and a third opening opposite to the second opening.
(33) In one embodiment, the accommodating cavity is configured to accommodate a 3D holographic device, the 3D holographic device comprising a 3D holographic device body extending into the accommodating cavity through the third opening and a 3D holographic device fixing surface, and the 3D holographic device fixing surface is connected to the first connecting surface.
(34) In one embodiment, the first reinforcing skeleton comprises two or more first reinforcing beams, and two ends of each of the first reinforcing beams are connected to the upper base frame.
(35) In one embodiment, the upper base body further comprises a second reinforcing skeleton, wherein the second reinforcing skeleton comprises two or more second reinforcing beams, and two ends of each of the second reinforcing beams are respectively connected to the upper base frame and the first reinforcing beam.
(36) In one embodiment, the first connecting member comprises grooves that match the first reinforcing skeleton.
(37) In one embodiment, when the first reinforcing skeleton comprises two of the first reinforcing beams, the first connecting member comprises two grooves, and each groove is equipped with one of the first reinforcing beams.
(38) In one embodiment, the first connecting member is provided with a first limiting surface opposite to the first connecting surface, and the second opening is provided on the first limiting surface.
(39) In one embodiment, the upper base further comprises a transparent plate that matches the first

opening, and the transparent plate abuts the first limiting surface.
(40) In one embodiment, the rotatable camera mounting device further comprises a lower base opposite to the upper base and a connecting unit connecting the upper base and the lower base.
(41) In one embodiment, the connecting unit comprises a second connecting member connected to the first connecting member, a bearing member connected to the second connecting member, and a rotatable disc fixing member connected to the bearing member and the lower base.
(42) In one embodiment, the second connecting member is provided with a second connecting surface near the first connecting surface, and the first connecting surface is connected to the second connecting surface.
(43) In one embodiment, the second connecting member is provided with a second connecting end away from the first connecting surface, and the second connecting end is connected to an upper end surface of a bearing inner ring of the bearing member.
(44) In one embodiment, the rotatable disc fixing member is provided with a third connecting surface near one end of the bearing member, and the third connecting surface is connected to a lower end surface of the bearing inner ring of the bearing member.
(45) In one embodiment, a third connecting end is provided at one end of the rotatable disc fixing member away from the bearing member, and the third connecting end is connected to the lower base.
(46) In one embodiment, the lower base is provided with a lower base frame and a third reinforcing skeleton, and the third reinforcing skeleton intersects with each other to form a fourth connecting end for connecting the rotatable disc fixing member.
(47) In one embodiment, the rotatable camera mounting device further comprises a driving device, the driving device comprising a motor, a driving wheel connected to the motor, the bearing member, and a rotating device connected to the bearing member, wherein the bearing member rotates with the driving wheel.
(48) In one embodiment, the bearing member further comprises a bearing outer ring, wherein the bearing outer ring is provided with teeth that match the driving wheel, and one end of the rotating device is connected to the bearing outer ring.
(49) In one embodiment, the rotating device comprises a first rotating unit connected to the bearing outer ring, a second rotating unit connected to the first rotating unit, and a third rotating unit rotatably connected to the second rotating unit.
(50) In one embodiment, the second rotating unit is nested inside the first rotating unit.
(51) In one embodiment, fasteners are provided on the first rotating unit to adjust length of the second rotating unit nested inside the first rotating unit and to fix the second rotating unit.
(52) In one embodiment, the side of the upper base frame has a channel, which is provided with an ambient light, and a transparent cover is provided on the outer side of the channel to adhere to the upper base frame.
(53) In one embodiment, the side of the lower base frame has a channel, which is provided with an ambient light, and the outer side of the channel is provided with a transparent cover that adheres to the lower base frame.
(54) In one embodiment, the rotatable camera mounting device further comprises an AI voice device and a main control device, wherein the main control device is configured to create instructions and control the rotatable camera mounting device to perform actions after obtaining user voice collected by the AI voice device.
(55) In one embodiment, the AI voice device comprises a microphone device configured to capture the user's voice when the user speaks.
(56) In one embodiment, the AI voice device further comprises a speaker device configured to provide voice feedback under the control of the main control device when the microphone device acquires user voice.
(57) In one embodiment, the action includes controlling activation, deactivation, lighting mode

switching, or brightness adjustment of the ambient light.
(58) In one embodiment, one end of the first rotating unit connected to the bearing outer ring is provided with multiple connecting fins, which extend outward to form a clearance space for avoiding the second connecting end.
(59) In one embodiment, the first rotating unit comprises multiple first cantilevers, and the second rotating unit comprises second cantilevers corresponding to the first cantilevers, each of which is nested inside the first cantilever.
(60) In one embodiment, the loading surface is detachably connected to the upper base frame and/or the loading surface is integrally formed with the upper base frame.
(61) According to another aspect of the present disclosure, there is provided a rotatable camera mounting device, comprising: a driving device comprising a motor, a rotatable disc device connected to the motor with a driving wheel, and a rotating device connected to the rotatable disc device, wherein the rotating device is configured to rotate circumferentially with the motor; and a rotatable disc fixing member used to fix the rotatable disc device on the rotatable camera mounting device.
(62) In one embodiment, the rotatable disk device comprises a bearing inner ring fixed to the rotatable camera mounting device by the rotatable disc fixing member, and a bearing outer ring that rotates relative to the bearing inner ring and engages with the driving wheel.
(63) In one embodiment, the rotatable camera mounting device comprises a lower base, which is provided with a lower base frame and a third reinforcing skeleton, and the rotatable disc fixing member is fixedly installed on the third reinforcing skeleton.
(64) In one embodiment, the rotatable camera mounting device further comprises: an upper base arranged opposite to the lower base, wherein, the upper base comprises a loading surface and an upper base body, the upper base body comprises an upper base frame and a first connecting member, the first connecting member is provided with a first connecting surface at one end away from the loading surface.
(65) In one embodiment, the rotatable camera mounting device further comprises a second connecting member connected to the first connecting surface and connected to the bearing inner ring.
(66) In one embodiment, the second connecting member is provided with a second connecting surface near the first connecting surface, and the first connecting surface is connected to the second connecting surface.
(67) In one embodiment, the second connecting member is provided with a second connecting end away from the first connecting surface, and the second connecting end is connected to an upper end surface of the bearing inner ring.
(68) In one embodiment, a third connecting surface is provided at one end of the rotatable disc fixing member near the bearing inner ring, and the third connecting surface is connected to a lower end surface of the bearing inner ring.
(69) In one embodiment, a third connecting end is provided at one end of the rotatable disc fixing member away from the bearing inner ring, and the third connecting end is connected to the third reinforcing skeleton.
(70) In one embodiment, the third reinforcing skeleton intersects with each other to form a fourth connecting end for connecting the third connecting end.
(71) In one embodiment, the rotating device comprises a first rotating unit connected to the bearing outer ring, a second rotating unit connected to the first rotating unit, and a third rotating unit rotatably connected to the second rotating unit.
(72) The rotatable camera mounting device can be equipped with a 3D holographic device, which can display contents such as time, pictures, animations, etc., further enhancing the entertainment; a stable skeletal structure, including multiple reinforcing beams, greatly enhancing the strength of the upper base; LED ambient light, which enhances the fun and playability of the device; an AI voice

control unit, greatly improving the efficiency and scene application of the device by adjusting the rotation of the device, switching or changing lights, and turning on and off the device by voice, enhancing the interaction between people and the device; the main electrical components such as the control device and motor are all located on the lower base, which can be installed and removed without disassembling the upper base, making it easy to install and maintain the entire machine.
(73) More embodiments of the present disclosure can also achieve other advantageous technical effects not listed one by one, which may be partially described in the following and can be expected and understood by those skilled in the art after reading the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In order to provide a clearer explanation of the technical solution in the embodiments of the present application, a brief introduction will be given below to the accompanying drawings required in the embodiments or prior art descriptions. It is evident that the accompanying drawings in the following description are only some embodiments of the present application. For those skilled in the art, other accompanying drawings can be obtained based on these drawings without the need for creative labor.
(2) FIG. **1** shows an exploded view of the rotatable camera mounting device of the present application;
(3) FIG. **2** shows an exploded view of the upper base of the rotatable camera mounting device of the present application;
(4) FIG. **3** shows an exploded view of the upper base, lower base, and the connecting unit of the rotatable camera mounting device of the present application;
(5) FIG. **4** shows a perspective view of the first connecting member of the rotatable camera mounting device of the present application;
(6) FIG. **5** shows another perspective view of the first connecting member of FIG. **4**;
(7) FIG. **6** shows another exploded view of the rotatable camera mounting device of the present application; and
(8) FIG. **7** shows the structure between the driving wheel and the bearing outer ring in one embodiment of the present application.
DETAILED DESCRIPTION
(9) In order to clarify the purpose, technical solution, and advantages of the embodiments of the present disclosure, the following will provide a clear and complete description of the technical solution in the embodiments of the present disclosure in conjunction with the accompanying drawings. Obviously, the described embodiments are one part of the embodiments of the present disclosure, not all of them. The components of the embodiments of the present disclosure described and illustrated in the accompanying drawings can be arranged and designed in various different configurations.
(10) Therefore, the detailed description of the embodiments of the present disclosure provided in the accompanying drawings is not intended to limit the scope of the present disclosure, but only to represent selected embodiments of the present disclosure. Based on the embodiments in the present disclosure, all other embodiments obtained by ordinary skilled persons in this field without creative labor are within the scope of protection of the present disclosure.
(11) It should be noted that similar labels and letters represent similar items in the following figures. Therefore, once an item is defined in one figure, it does not need to be further defined and explained in subsequent figures.
(12) It should be understood that the terms “up”, “down”, “front”, “back”, “left”, “right”, “horizontal”, etc. indicate the orientation or position relationship based on the orientation or

position relationship shown in the drawings, only for the convenience of describing and simplifying the description of the present application, rather than indicating or implying that the device or component referred to must have a specific orientation, be constructed and operated in a specific orientation, and therefore cannot be understood as a limitation of the present application. In addition, the terms “first”, “second”, “third”, etc. are only used to distinguish descriptions and cannot be understood as indicating or implying relative importance. In the description of the present disclosure, unless otherwise specified, the meaning of “multiple” refers to two or more.

(13) As used in this application, the terms “installation”, “connection”, “connecting”, “fixed” and other terms should be broadly understood, for example, they can be fixed connections, detachable connections, or integrated connections; It can be a mechanical connection or an electrical connection; It can be directly connected, or indirectly connected through an intermediate medium, or it can be an internal connection between two components. For ordinary technical personnel in this field, the specific meanings of the above terms in this application can be understood based on specific circumstances.

(14) In the present disclosure, unless otherwise specified and limited, the first feature above or below the second feature may include direct contact between the first and second features, or may include contact between the first and second features through another feature between them instead of direct contact. Moreover, the first feature being “above”, and “on” the second feature includes the first feature being directly above and diagonally above the second feature, or simply indicating that the first feature is horizontally higher than the second feature. The first feature being “below” include the first feature being directly below and diagonally below the second feature, or simply indicating that the first feature has a lower horizontal height than the second feature.

(15) The embodiments of the present disclosure are described in detail below, and examples of the embodiments are shown in the accompanying drawings, where the same or similar reference numerals from beginning to end represent the same or similar elements or elements with the same or similar functions. The embodiments described below with reference to the accompanying drawings are exemplary and are only used to explain the present disclosure, and should not be understood as limiting the present disclosure.

(16) As shown in FIGS. **1-5**, the rotatable camera mounting device with a rotatable disc device comprises: an upper base **1**, wherein the upper base **1** comprises a loading surface **10**, an upper base body **11** connected to the loading surface **10**, the upper base body **11** comprising an upper base frame **110**, a first reinforcing skeleton **111** connected to the upper base frame **110**, and a first connecting member **112**, wherein the first connecting member **112** is provided with an accommodating cavity **1120**, and a first connecting surface **1121** is provided at one end of the first connecting member **112** away from the loading surface **10**. As shown in FIG. **6**, the loading surface **10** is provided with a first opening **101**. The upper base **1** also includes a transparent plate **12** that matches the first opening **101**, such as a transparent tempered glass, whose surface is flush with the surface of the loading surface **10**. The transparent tempered glass, due to its high hardness and wear resistance, can ensure that people or objects are placed on it without damage. The loading surface **10** can be used for standing people or placing items. The loading surface **10** is detachably connected to the upper base frame **110** and/or the loading surface **10** is integrally formed with the upper base frame **110**.

(17) As shown in FIGS. **4-5**, the first connecting member **112** is provided with a second opening **1122** opposite to the first opening **101** and a third opening **1123** opposite to the second opening **1122**. As shown in FIGS. **2**, **4**, and **5**, the accommodating cavity **1120** is configured to accommodate the 3D holographic device **2**. The 3D holographic device **2** includes a 3D holographic device body **20** that extends into the accommodating cavity **1120** through a third opening **1123**, and a 3D holographic device fixing surface **21** that is connected to the first connecting surface **1121**. In one embodiment, holes are provided on the fixing surface **21** of the 3D holographic device, and the first connecting surface **1121** is also provided with holes. The fixing

surface **21** of the 3D holographic device is connected to the first connecting surface **1121** by installing screws on the holes. The transparent tempered glass is used to display the program images of 3D holographic device **2**. After the 3D holographic device **2** is activated, it can display animations, images, time, etc. through the transparent properties of the tempered glass.
(18) As shown in FIG. **2**, the first reinforcing skeleton **111** includes two or more first reinforcing beams **1110**, and two ends of each of the first reinforcing beams **1110** are connected to the upper base frame **110**. In one embodiment, the first reinforcing skeleton **111** comprises two first reinforcing beams **1110**. The two ends of each of the first reinforcing beams **1110** are connected to the upper base frame **110**, and the two first reinforcing beams **1110** extend parallel to each other or in approximately the same direction. By setting the first reinforcing beam **1110** to enhance the strength of the upper base frame **110**, the loading surface **10** can be used for standing people or placing objects.
(19) As shown in FIG. **2**, the upper base body **11** also includes a second reinforcing skeleton **113**, which includes two or more second reinforcing beams **1130**. The two ends of each of the second reinforcing beams **1130** are respectively connected to the upper base frame **110** and the first reinforcing beam **1110**. In one embodiment, the second reinforcing skeleton **113** includes four second reinforcing beams **1130**, and two ends of each of the second reinforcing beams **1130** are respectively connected to the upper base frame **110** and the first reinforcing beam **1110**. The length of the second reinforcing beam **1130** is shorter than that of the first reinforcing beam **1110**. Each of the first reinforcing beams **1110** is connected to two of the second reinforcing beams **1130**. By setting the second reinforcing beam **1130**, the strength of the upper base frame **110** can be further enhanced, so that the loading surface **10** can be used for standing people or placing objects. Due to the need to carry people or objects for photography, the load-bearing and stability of the upper base **1** have been further strengthened structurally with multiple reinforcing beams, greatly enhancing the load-bearing strength of the upper base **1**.
(20) As shown in FIG. **2**, the two ends of each of the first reinforcing beam **1110** are connected to the upper base frame **110**, while two ends of each of the second reinforcing beams **1130** are respectively connected to the upper base frame **110** and the first reinforcing beam **1110**. In one embodiment, holes can be provided on the first connecting member **112**, and holes can also be provided at corresponding positions of the first reinforcing beam **1110**. The first connecting member **112** is fixedly connected to the first reinforcing beam **1110** by screws and the likes. Alternatively, the first connecting member **112** can be fixedly connected to the first reinforcing beam **1110** through welding.
(21) As shown in FIGS. **4-5**, the first connecting member **112** also includes grooves **1124** that match the first reinforcing skeleton **111**. When the first reinforcing skeleton **111** includes two first reinforcing beams **1110**, the first connecting member **112** includes two grooves **1124**, each groove **1124** is equipped with one first reinforcing beam **1110**. In one embodiment, two first reinforcing beams **1110** extend parallel to each other or in approximately the same direction, and each groove **1124** is equipped with a first reinforcing beam **1110**, restricting the left and right shaking or moving of the first connecting member **112** through the first reinforcing beam **1110**. The first connecting member **112** and the first reinforcing beam **1110** can be connected by screws. Alternatively, the first connecting member **112** and the first reinforcing beam **1110** can be connected by welding.
(22) As shown in FIGS. **4-5**, the first connecting member **112** is provided with a first limiting surface **1125** opposite to the first connecting surface **1121**, and the second opening **1122** is provided on the first limiting surface **1125**. In one embodiment, when the upper base **1** includes a transparent plate **12** that matches the first opening **101**, the transparent plate **12** contacts the first limiting surface **1125**. The transparent plate **12** is set inside the first opening **101**, which means that the outer side of the transparent plate **12** is in contact with the inner side of the first opening **101**, so the left and right movement of the transparent plate **12** is restricted by the first opening **101**. By setting the first limiting surface **1125**, the movement of the transparent plate **12** in the height direction can

be restricted.

(23) As shown in FIGS. **1-3** and **6**, the rotatable camera mounting device with a rotatable disc device also includes a lower base **3** opposite to the upper base **1** and a connecting unit **4** connecting the upper base **1** and the lower base **3**. The connecting unit **4** includes a second connecting member **41** connected to the first connecting member **112**, a rotatable disc device **42** connected to the second connecting member **41**, and a rotatable disc fixing member **43** connected to the rotatable disc device **42** and the lower base **3**. The rotatable disc fixing member **43** is used for fixing the rotatable disc device **42** on the rotatable camera mounting device. The second connecting member **41** is provided with a second connecting surface **410** near the first connecting surface **1121**, and the first connecting surface **1121** is detachably connected to the second connecting surface **410**. The second connecting member **41** is provided with a second connecting end **411** away from the first connecting surface **1121**, and the second connecting end **411** is connected to an upper end surface of a bearing inner ring **420** of the rotatable disc device **42**. A third connecting surface **430** is provided at one end of the rotatable disc fixing member **43** near the rotatable disc device **42**, and the third connecting surface **430** is connected to the lower end surface of the bearing inner ring **420** of the rotatable disc device **42**. The end of the rotatable disc fixing member **43** away from the rotatable disc device **42** is provided with a third connecting end (not shown), which is connected to the lower base **3**. The lower end of the lower base **3** is equipped with foot cups **30**, which can be adjusted to a certain height, allowing the rotatable camera mounting device to remain stable in different usage environments.

(24) As shown in FIGS. **1**, **3**, and **6**, the lower base **3** is equipped with a lower base frame **31** and a third reinforcing skeleton **32**. The third reinforcing skeleton **32** intersects with each other to form a fourth connecting end **320** for connecting the third connecting end.

(25) As shown in FIGS. **1**, **3**, and **6**, the rotatable camera mounting device also includes a driving device **5**, which includes a motor **50**, a driving wheel **51** connected to the motor **50**, a rotatable disc device **42**, and a rotating device **52** connected to the rotatable disc device **42**. The rotatable disc device **42** engages with the driving wheel **51**. The rotating device **52** can rotate in the circumferential direction around the upper base **1** and the lower base **3** and can be equipped with a camera equipment. The camera equipment can include mobile phones, tablets, computers, cameras, etc. Motor **50** can be used to drive the rotating device **52** to rotate in the circumferential direction around the upper base **1** and the lower base **3**. In one embodiment, the rotatable disc device **42** can rotate with the driving wheel **51** under the action of a belt.

(26) As shown in FIG. **3**, the rotatable disc device **42** also includes a bearing inner ring **420** fixed on the rotatable camera mounting device by a rotatable disc fixing member **43**, and a bearing outer ring **421** that can be rotated relative to the bearing inner ring **420**. The bearing outer ring **421** is provided with teeth that match the driving wheel **51**, and one end of the rotating device **52** is connected to the bearing outer ring **421**. The rotating device **52** is configured to rotate circumferentially under the drive of the motor **50**. The rotatable disc device **42** of the driving device **5** is installed at the upper end surface of the lower base **3**. The rotatable disc device **42** is almost a specially designed bearing, which is divided into a bearing inner ring **420** and a bearing outer ring **421**. Both the bearing inner ring **420** and the bearing outer ring **421** can easily rotate. The rotatable disc device **42** is fixedly installed on the lower base **3** through the holes on the bearing inner ring **420**, ensuring that the bearing inner ring **420** of the rotatable disc device **42** remains fixed. The bearing outer ring **421** of the rotatable disc device **42** is provided with teeth and mounting holes for the rotating device **52**, making it easy for the rotating device **52** to be fixedly installed on the outside of the rotatable disc device **42**. In addition, the motor **50** is fixedly installed at the bottom of the lower base **3**. After installation, the driving wheel **51** connected to the motor **50** engages perfectly with the teeth of the bearing outer ring **421**. When the motor **50** is working, the rotation of the small gear or wheel on the motor **50** drives the large gear or wheel on the outside of the rotatable disc device **42** to rotate together, further driving the rotating device **52** fixedly

installed on the bearing outer ring **421** to rotate in the circumferential direction around the upper base **1** and the lower base **3**. In one embodiment, the bearing outer ring **421** can rotate with the driving wheel **51** under the action of the belt, as shown in FIG. **7**. In one embodiment, the bearing outer ring **421** can also be connected to another gear to act as a driven wheel.

(27) As shown in FIGS. **1** and **3**, the rotating device **52** includes a first rotating unit **520** connected to the bearing outer ring **421**, a second rotating unit **521** connected to the first rotating unit **520**, and a third rotating unit **522** rotatably connected to the second rotating unit **521**. The second rotating unit **521** are nested inside the first rotating unit **520**. The third rotating unit **522** can be set as a telescopic rod, used to fix various auxiliary equipment such as fill lights, mobile phones, cameras, tablets, etc. The rod body and its connecting members are made of metal material, which enhances the stability and durability of the product.

(28) As shown in FIGS. **1** and **3**, the first rotating unit **520** is equipped with fasteners **5200**, which are used to adjust length of the second rotating unit **521** nested inside the first rotating unit **520** and to fix the second rotating unit **521**. The fasteners **5200** can be screws, for example. The second rotating unit **521** can be adjusted by pulling and pushing on the first rotating unit **520**, and then fixed by screws, making it easy to adjust the distance between the telescopic rod and the upper base **1** and lower base **3** as needed, allowing users to expand more photography application scenarios.

(29) As shown in FIGS. **1** and **3**, one end of the first rotating unit **520** connected to the bearing outer ring **421** is provided with multiple connecting fins **5201**, which extend outward to form a clearance space for avoiding the second connecting end **411**. The first rotating unit **520** includes multiple first cantilevers **5202**, and the second rotating unit **521** includes second cantilevers **5210** corresponding to the first cantilevers **5202**, each of which is nested inside the first cantilever **5202**. The first cantilever **5202** is slidably connected to the second cantilever **5210**, and the second cantilever **5210** is installed on the first cantilever **5202** through the fasteners **5200**.

(30) As shown in FIG. **6**, the side of the upper base frame **110** has a channel G, which is equipped with ambient lights L, such as LED lights. The outer side of the channel G is equipped with a transparent cover C that adheres to the upper base frame **110**. Similarly, the side of the lower base frame **31** has a channel G, which is equipped with ambient lights L, such as LED lights. The outer side of the channel G is equipped with a transparent cover C that adheres to the lower base frame **31**. For example, the channel G and the ambient light L set inside the channel G can be covered with transparent acrylic to ensure that the LED light is not damaged due to collision during use or transportation, and to make the light softer, more uniform, and not glaring, enhancing the texture of the equipment.

(31) As shown in FIGS. **1** and **3**, the rotatable camera mounting device also includes an AI voice device **6** and a main control device **7**. The main control device **7** is configured to create instructions and control the rotatable camera mounting device to perform actions after obtaining user voice collected by the AI voice device **6**. The main control device **7** can be used to control the rotation of the motor. The main control device **7** can control the turn on, turn off, light mode switching, or brightness adjustment of the ambient lighting. The AI voice device **6** includes a microphone device configured to capture the user's voice when they speak. User voice can include words, sentences, songs, etc. The AI voice device **6** also includes a speaker device, which is configured to provide voice feedback under the control of the main control device **7** when the microphone device acquires user voice. The actions include controlling the activation, deactivation, lighting mode switching, or brightness adjustment of ambient lighting. It should be further explained that both the AI voice device **6** and the driving device are installed on the side of the lower base near the foot cup, which is more convenient for installation, disassembly, maintenance, etc.

(32) The rotatable camera mounting device can be equipped with a 3D holographic device, which can display contents such as time, pictures, animations, etc., further enhancing the entertainment; a stable skeletal structure, including multiple reinforcing beams, greatly enhancing the strength of the upper base; LED ambient light, which enhances the fun and playability of the device; an AI voice

control unit, greatly improving the efficiency and scene application of the device by adjusting the rotation of the device, switching or changing lights, and turning on and off the device by voice, enhancing the interaction between people and the device; the main electrical components such as the control device and motor are all located on the lower base, which can be installed and removed without disassembling the upper base, making it easy to install and maintain the entire machine.
(33) The above description is only the preferred embodiment of the present disclosure and is not intended to limit the present disclosure. Any modifications, equivalent substitutions, and improvements made within the spirit and principles of the present disclosure should be included in the scope of protection of the present disclosure.

Claims

1. A rotatable camera mounting device with a rotatable disc device, wherein comprising: a driving device comprising a motor, a rotatable disc device connected to the motor, and a rotating device connected to the rotatable disc device, wherein the rotating device is configured to rotate circumferentially with the motor; a rotatable disc fixing member used to fix the rotatable disc device on the rotatable camera mounting device; and an upper base, the upper base comprises an upper base body comprising an upper base frame and a first reinforcing skeleton connected to the upper base frame, the first reinforcing skeleton comprising two or more first reinforcing beams, and the two ends of each of the first reinforcing beams are connected to the upper base frame, and the upper base body further comprises a second reinforcing skeleton, wherein the second reinforcing skeleton comprises two or more second reinforcing beams, and the two ends of each of the second reinforcing beams are respectively connected to the upper base frame and the first reinforcing beams, and a lower base arranged opposite to the upper base, the lower base is provided with a lower base frame and a third reinforcing skeleton, and the rotatable disc fixing member is fixedly installed on the third reinforcing skeleton, wherein the rotatable disc device comprises a bearing inner ring fixed to the rotatable camera mounting device by the rotatable disc fixing member, and a bearing outer ring that rotates relative to the bearing inner ring, the driving device further comprises a driving wheel connected to the motor, and the bearing outer ring rotates with the driving wheel, the upper base further comprises a loading surface, the upper base body comprises a first connecting member, the first connecting member is provided with a first connecting surface at one end away from the loading surface, the first connecting member comprises grooves that match with the first reinforcing skeleton.
2. The rotatable camera mounting device with a rotatable disc device according to claim 1, wherein the rotatable camera mounting device further comprises a second connecting member connected to the first connecting surface and connected to the bearing inner ring.
3. The rotatable camera mounting device with a rotatable disc device according to claim 2, wherein the second connecting member is provided with a second connecting surface near the first connecting surface, and the first connecting surface is connected to the second connecting surface.
4. The rotatable camera mounting device with a rotatable disc device according to claim 3, wherein the second connecting member is provided with a second connecting end away from the first connecting surface, and the second connecting end is connected to an upper end surface of the bearing inner ring.
5. The rotatable camera mounting device with a rotatable disc device according to claim 4, wherein the rotatable disc fixing member is provided with a third connecting surface near one end of the bearing inner ring, and the third connecting surface is connected to a lower end surface of the bearing inner ring.
6. The rotatable camera mounting device with a rotatable disc device according to claim 5, wherein the third reinforcing skeleton intersects with each other to form a fourth connecting end for connecting the rotatable disc fixing member.

7. The rotatable camera mounting device with a rotatable disc device according to claim 6, wherein the rotating device comprises a first rotating unit connected to the bearing outer ring, a second rotating unit connected to the first rotating unit, and a third rotating unit rotatably connected to the second rotating unit.

8. The rotatable camera mounting device with a rotatable disc device according to claim 7, wherein the second rotating unit is nested inside the first rotating unit.

9. The rotatable camera mounting device with a rotatable disc device according to claim 8, wherein fasteners are provided on the first rotating unit to adjust length of the second rotating unit nested inside the first rotating unit and to fix the second rotating unit.

10. The rotatable camera mounting device with a rotatable disc device according to claim 9, wherein one end of the first rotating unit connected to the bearing outer ring is provided with multiple connecting wings, which extend outward to form a clearance space for avoiding the second connecting end.

11. The rotatable camera mounting device with a rotatable disc device according to claim 10, wherein the first rotating unit comprises multiple first cantilevers, the second rotating unit comprises second cantilevers corresponding to the first cantilevers, and each of the second cantilevers is nested inside the first cantilever.

12. The rotatable camera mounting device with a rotatable disc device according to claim 1, wherein the first connecting member is provided with a first limiting surface opposite to the first connecting surface.

13. The rotatable camera mounting device with a rotatable disc device according to claim 1, wherein the loading surface is provided with a first opening, the first connecting member is provided with a third opening, the rotatable camera mounting device further comprises a 3D holographic device, the 3D holographic device comprising a 3D holographic device body extending into an accommodating cavity through the third opening and a 3D holographic device fixing surface, and the accommodating cavity is enclosed by the first connecting member and the first reinforcing skeleton.